MATHEMATICS / INDEPENDENT STUDY

MM5020

Mathematics III - Abstract Algebra

Complete solutions

Original teaching material aligned with Stockholm University course descriptions.
Independently authored; not an official university publication.

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How to use this volume E problems practice definitions and standard arguments; C problems combine ideas; A problems require synthesis or generalization. These are study difficulty labels, not official SU grading criteria. Time estimates are guides, not deadlines. Write a complete attempt before opening hints. In the hints volume, read Hint 1 first and return to the problem before consulting Hint 2. Solutions give an argument and criteria for self-assessment. Expanded solutions also explain how to find the approach, a tempting dead end, and the role of the hypotheses.

Each unit has twelve core problems (four E, five C, three A) and an optional A-level transfer challenge numbered 13. The challenge asks you to choose a method in a new setting; budget its hour separately. Problem 12 and the challenge have detailed solutions with independently written decisive ideas. Mock estimates total 170 minutes of problem work plus ten minutes for review.

1 Unit 1 · Groups and cyclic structure

MM5020-U1.01 · E · 15 min
Learning objective: U1.1 Group axioms

Show that the even integers form a group under addition, and explain why they do not form a group under multiplication.

Sums and negatives of even integers are even; addition is associative and has identity zero. Thus they form an additive group. A multiplicative identity inside this set would satisfy $e \cdot 2 = 2$, forcing $e = 1$, which is not even. Hence multiplication does not give a group.

Check your argument. Check closure, associativity, identity, and inverse; a single failed multiplicative axiom suffices.

MM5020-U1.02 · E · 15 min
Learning objective: U1.1 Cancellation

Solve $axb = c$ in an arbitrary group and prove uniqueness.

The solution is $a^{-1}cb^{-1}$. Substitution gives $a(a^{-1}cb^{-1})b = c$. Any solution must have this value after the indicated multiplications, proving uniqueness.

Check your argument. Keep the factor order; do not commute arbitrary elements.

MM5020-U1.03 · E · 15 min
Learning objective: U1.2 Element orders

Find the order and generated subgroup of $[9]$ in $\mathbb{Z}/15\mathbb{Z}$.

The order is five. The successive multiples are $[0]$, $[9]$, $[3]$, $[12]$, $[6]$, then $[0]$. These five elements form the generated subgroup.

Check your argument. Give both the order and the actual residue classes.

MM5020-U1.04 · E · 15 min
Learning objective: U1.2 Subgroup test

Prove that the intersection of two subgroups of a group is a subgroup.

Let $H, K \leq G$. Their intersection contains e. For $a, b \in H \cap K$, both subgroups contain ab^{-1}, so it lies in the intersection. The subgroup test proves the claim.

Check your argument. Explain nonemptiness as well as closure under ab^{-1}.

MM5020-U1.05 · C · 30 min
Learning objective: U1.1 Structural identities

Prove that a group in which $x^2 = e$ for every x is abelian.

For any a, b, the hypothesis gives $a^{-1} = a$, $b^{-1} = b$, and $(ab)^{-1} = ab$. The inverse-of-product identity gives $ab = ba$. Thus the group is abelian.

Check your argument. Apply the hypothesis to the product as well as the individual factors.

MM5020-U1.06 · C · 30 min
Learning objective: U1.2 Cyclic subgroups

List all subgroups of $\mathbb{Z}/18\mathbb{Z}$ and their orders.

The subgroups are generated by $[1]$, $[2]$, $[3]$, $[6]$, $[9]$, $[18]$, with orders $18, 9, 6, 3, 2, 1$ respectively. Every subgroup is cyclic; choosing its least positive representative shows its generator divides eighteen, so the list is exhaustive.

Check your argument. Justify completeness, not only the six constructions.

MM5020-U1.07 · C · 30 min
Learning objective: U1.2 Generator arithmetic

Let a have order n. Prove $\langle a^r \rangle = \langle a^{\gcd(n,r)} \rangle$.

Write $d = \gcd(n, r)$. Since d divides r, a^r is a power of a^d, giving one inclusion. Bézout gives $d = un + vr$, so $a^d = a^{un}(a^r)^v = (a^r)^v$, giving the reverse inclusion.

Check your argument. Prove both inclusions and use $a^n = e$.

MM5020-U1.08 · C · 30 min
Learning objective: U1.2 Subgroup structure

Prove that the union of two subgroups H, K is a subgroup exactly when one contains the other.

If one subgroup contains the other, their union is that subgroup. Otherwise choose $h \in H \setminus K$ and $k \in K \setminus H$. If $hk \in H$, then $k = h^{-1}(hk) \in H$, a contradiction. If $hk \in K$, then $h = (hk)k^{-1} \in K$, also a contradiction. Thus the union is not closed.

Check your argument. Use the correct multiplication side in each contradiction.

MM5020-U1.09 · C · 30 min
Learning objective: U1.1 Noncommutative examples

Let r rotate a square by a quarter turn and s reflect it in a vertical axis. Show $srs = r^{-1}$ and use this to prove the symmetry group is nonabelian.

Label vertices cyclically by residues modulo four so that $r(i) = i + 1$ and write the reflection as $s(i) = c - i$ modulo four for a fixed residue c. Then $srs(i) = c - (c - i + 1) = i - 1 = r^{-1}(i)$. If r, s commuted, this would give $r = r^{-1}$, contradicting the order four of r.

Check your argument. Give a precise labeling or coordinate calculation.

MM5020-U1.10 · A · 50 min
Learning objective: U1.2 Automorphisms

Classify all automorphisms of the additive group $\mathbb{Z}/n\mathbb{Z}$.

Every endomorphism is $f_{k([r])} = [kr]$, since additivity determines all values from $[1]$. This is well defined modulo n. Its image is the subgroup generated by $[k]$, of order $\frac{n}{\gcd(n,k)}$. Thus it is bijective exactly when $\gcd(n, k) = 1$. The automorphisms are precisely these multiplication maps; composition corresponds to multiplication of unit residues modulo n.

Check your argument. Prove that every map is accounted for and determine the exact invertibility condition.

MM5020-U1.11 · A · 50 min
Learning objective: U1.2 Products of cyclic groups

Prove that $C_m \times C_n$ is cyclic exactly when $\gcd(m, n) = 1$.

For any pair (x, y), its order divides $\operatorname{lcm}(m, n)$ because that exponent kills both coordinates. If the gcd exceeds one, this least common multiple is smaller than mn, so no element generates the product. If the gcd is one, a pair of generators has order exactly $\operatorname{lcm}(m, n) = mn$: an exponent killing both is divisible by both m, n. Thus this pair generates.

Check your argument. Prove impossibility in the noncoprime case for every pair, not just one candidate.

MM5020-U1.12 · A · 50 min
Learning objective: U1.1 Axioms and finiteness

Let M be a finite nonempty set with an associative operation and a two-sided identity. Assume left and right cancellation. Prove that M is a group.

Decisive idea. Convert algebraic cancellation into a finite-set argument, then reconcile the two one-sided inverse witnesses.

Finding the approach. An inverse is a solution to an equation. Translate the question about a into the two maps given by multiplication, so that cancellation becomes injectivity.

Argument. For $a \in M$, both maps $x \mapsto ax$ and $x \mapsto xa$ are injective by cancellation and hence surjective. Choose b, c with $ab = e$ and $ca = e$. Associativity gives $c = c(ab) = (ca)b = b$, so b is a two-sided inverse. Every element has an inverse and the other group axioms were assumed.

More explicitly, injectivity of $x \mapsto ax$ follows from $ax = ay \Rightarrow x = y$. On a finite set an injective self-map has an image with the same cardinality as the set, so it hits e. Repeat on the other side. The equality $c = b$ is the step that turns two potentially different one-sided inverses into one inverse.

A tempting dead end. Do not cancel a factor in an equation you have not yet constructed, or assume that the powers of a already include an inverse. Infinite cancellative monoids, such as nonnegative integers under addition, need not be groups.

Hypotheses. Finiteness supplies surjectivity, cancellation supplies injectivity, associativity joins the two inverses, and the identity lets us formulate their equations.

Check your argument. Identify precisely where finiteness and associativity enter.

MM5020-U1.13 · A · 60 min · Optional challenge

Learning objective: U1 Transfer: A disguised operation

On $X = \mathbb{R}\{-1\}$ define $a \star b = a + b + ab$. Determine whether this is a group, find its identity and inverses, and find all elements of finite order.

Decisive idea. A change of coordinates can transport all the group axioms at once, while still requiring an explicit bijection and compatibility check.

Finding the approach. The quadratic term obscures a familiar operation. Adding one removes the extra linear terms in a single factorization.

Argument. The map $\varphi : X \to \mathbb{R}\{0\}$, $\varphi(a) = 1 + a$, is a bijection and satisfies $\varphi(a \star b) = \varphi(a)\varphi(b)$. Closure follows because a product of nonzero reals is nonzero. Associativity and commutativity follow by applying the injective map φ to the corresponding equations. The identity has image one, so is 0. The inverse of a has image $(1 + a)^{-1}$, hence is $-\frac{a}{1+a}$. If a has finite order, $(1 + a)^n = 1$ for some positive integer n. A positive real root of unity is one and a negative real root of unity is minus one, so $a = 0$ or $a = -2$. These have orders one and two.

A tempting dead end. Checking a few numerical triples cannot prove associativity. Expanding both sides works but conceals the reason the operation has all the group properties.

Hypotheses. Removing minus one is essential: it corresponds to removing zero from multiplication. The classification of finite-order elements uses the real numbers; over the complex numbers there are many more roots of unity.

Check your argument. Verify closure, the inverse formula, and both directions of the finite-order classification.

2 Unit 2 · Cosets and permutations

MM5020-U2.01 · E · 15 min
Learning objective: U2.1 Cosets
List the cosets of $H = \{[0], [4], [8]\}$ in $\mathbb{Z}/12\mathbb{Z}$.
They are $\{0, 4, 8\}$, $\{1, 5, 9\}$, $\{2, 6, 10\}$, and $\{3, 7, 11\}$, with entries understood modulo twelve. They are disjoint and cover all twelve residues.
Check your argument. List each coset once and account for all elements.

MM5020-U2.02 · E · 15 min
Learning objective: U2.1 Lagrange
What element orders are possible in a group of order twenty-one according to Lagrange's theorem? Does this alone guarantee an element of order twenty-one?
The possible orders permitted by Lagrange are $1, 3, 7, 21$. Lagrange gives no existence conclusion for order twenty-one; it only says that an element's order, if it occurs, must divide twenty-one.
Check your argument. Distinguish an allowed order from a guaranteed order.

MM5020-U2.03 · E · 15 min
Learning objective: U2.2 Permutation composition
Compute $(123)(23)$ and its sign, composing right to left.
The product is (12). Its sign is negative. Independently, the three-cycle is even and the transposition is odd, so the product is odd.
Check your argument. State the composition convention and verify images.

MM5020-U2.04 · E · 15 min
Learning objective: U2.2 Cycle order
Find the order and sign of $(1234)(567)$ in S_7.
The order is $\operatorname{lcm}(4, 3) = 12$. The four-cycle has sign $(-1)^3 = -1$ and the three-cycle sign $(-1)^2 = 1$, so the product has sign -1.
Check your argument. Do not add cycle lengths when computing order.

MM5020-U2.05 · C · 30 min
Learning objective: U2.1 Coset criteria
Prove $aH = bH$ if and only if $a^{-1}b \in H$.
If the cosets agree, $b \in aH$, so $b = ah$ and $a^{-1}b = h \in H$. Conversely, if $b = ah$ with $h \in H$, then $bH = ahH = aH$ because multiplication by h permutes H.
Check your argument. Give both implications; either inverse orientation is valid if used consistently.

MM5020-U2.06 · C · 30 min
Learning objective: U2.1 Prime order
Prove that a group of prime order has no proper nontrivial subgroup and is cyclic.
Any subgroup has order dividing the prime p, hence order one or p. For $g \neq e$, $\langle g \rangle$ is not the trivial subgroup, so it has order p and equals the whole group.
Check your argument. Explain the existence of a nonidentity element and why its generated subgroup is nontrivial.

MM5020-U2.07 · C · 30 min
Learning objective: U2.2 Powers of cycles
Describe the disjoint cycles of $(123456)^2$ and $(123456)^3$, and compute their orders.
The square is $(135)(246)$ and has order three. The cube is $(14)(25)(36)$ and has order two. These agree with $\frac{6}{\gcd(6,2)}$ and $\frac{6}{\gcd(6,3)}$.
Check your argument. Show the actual cycles, not only the order formula.

MM5020-U2.08 · C · 30 min
Learning objective: U2.2 Counting permutations
How many three-cycles are in S_5? How many products of two disjoint transpositions?
There are $\binom{5}{3} \cdot 2 = 20$ three-cycles. For double transpositions choose the omitted fixed point in five ways, then one of three pairings of the remaining four points, giving fifteen.

Check your argument. Avoid counting rotations of a cycle or orders of disjoint factors as different permutations.

MM5020-U2.09 · C · 30 min
Learning objective: U2.2 Alternating groups

Show every three-cycle is even and every even permutation is a product of three-cycles.

A three-cycle is a product of two transpositions, so is even. Pair transpositions in an even factorization. Equal pairs cancel; pairs sharing one letter form a three-cycle or its inverse. For disjoint transpositions, $(ab)(cd) = [(ab)(bc)][(bc)(cd)]$, since the middle transpositions cancel. Each bracket is a three-cycle. Thus every pair can be replaced by three-cycles.

Check your argument. Handle equal, overlapping, and disjoint pairs.

MM5020-U2.10 · A · 50 min
Learning objective: U2.2 Minimal permutation degree

Find the least n for which S_n has an element of order fifteen.

Disjoint cycles of lengths three and five give order fifteen on eight letters. If fewer than eight letters are used, no cycle can have length fifteen. To make the least common multiple divisible by both three and five, there must therefore be distinct cycles with lengths divisible by three and five; these use at least eight letters. Hence the least degree is eight.

Check your argument. Prove the lower bound, not merely exhibit an example.

MM5020-U2.11 · A · 50 min
Learning objective: U2.1 Indices in a tower

For finite G and $K \leq H \leq G$, prove $[G : K] = [G : H][H : K]$ by constructing coset representatives.

Let g_i represent G/H and h_j represent H/K. Every g is $g_i h$ and then $g_i h_j k$, so the products represent all cosets modulo K. If $g_i h_j K = g_{i'} h_{j'} K$, then the elements lie in the same coset modulo H, forcing $i = i'$. Cancelling g_i forces $h_j K = h_{j'} K$ and hence $j = j'$. Count the products.

Check your argument. Establish both coverage and uniqueness.

MM5020-U2.12 · A · 50 min
Learning objective: U2.2 Conjugation and cycles

Prove $\tau(a_1 \ldots a_r)\tau^{-1} = (\tau(a_1) \ldots \tau(a_r))$. Deduce that conjugate permutations have the same cycle lengths, and conversely.

Decisive idea. Conjugation changes the names of points while preserving the directed cyclic order.

Finding the approach. A permutation identity is best tested at individual points. Choose a point whose name makes the rightmost map easy to evaluate.

Argument. The conjugate sends $\tau(a_i)$ to $\tau(a_{i+1})$ cyclically and fixes points outside the displayed support. Thus conjugation relabels each cycle and preserves lengths. Conversely, if two permutations have matching cycle lengths, define τ by matching the ordered entries of corresponding cycles, including fixed points. The displayed formula makes the conjugate equal to the target permutation.

At $\tau(a_i)$ the three successive images are a_i, a_{i+1}, and $\tau(a_{i+1})$. An outside point is sent by τ^{-1} outside the original support and hence returns unchanged. For the converse, pairing entire ordered cycles defines a bijection because disjoint cycles partition the set. Include cycles of length one so that this bijection is defined everywhere.

A tempting dead end. Matching only one point from each cycle leaves most of τ undefined. Equality of the number of cycles alone also does not match their lengths.

Hypotheses. This is a statement in the full symmetric group. A conjugating permutation constructed this way need not be even, so the same converse cannot be asserted without qualification in an alternating group.

Check your argument. Include fixed points and verify that the matching defines a bijection.

MM5020-U2.13 · A · 60 min · Optional challenge
Learning objective: U2 Transfer: Symmetries with fixed positions

How many permutations of six letters have order two and exactly two fixed letters? Prove that any two are conjugate, and determine how many permutations commute with one of them.

Decisive idea. Cycle structure both counts the objects and identifies the orbit whose stabilizer is being asked for.

Finding the approach. Start with the disjoint-cycle description, because order and fixed points are both visible there. Once the collection is one conjugacy class, its size determines a centralizer.
Argument. There are $\binom{6}{2} = 15$ choices of fixed letters. For the other four, match the first chosen letter with one of the other three; the remaining pair is forced. Thus there are 45 permutations. Each has type $(ab)(cd)$ with two fixed points, so relabeling the two pairs and the fixed points gives a conjugating permutation between any two. The conjugacy orbit has size 45. Its stabilizer is the centralizer, so orbit–stabilizer gives $\frac{6!}{45} = 16$. This last count can also be checked directly: independently flip the two moving pairs, interchange those pairs, and permute the two fixed letters, giving $2^2 \cdot 2 \cdot 2 = 16$.
A tempting dead end. Choosing two ordered pairs and then multiplying counts orders and orientations that do not change the permutation.
Hypotheses. The ambient group is the full symmetric group on six letters. Exactly two fixed points excludes the other involution types.
Check your argument. Explain the factor three in the pairing count and justify why the commuting permutations form the stabilizer.

3 Unit 3 · Homomorphisms and quotients

MM5020-U3.01 · E · 15 min
Learning objective: U3.1 Kernels
Find the kernel and image of $f : \mathbb{Z} \to \mathbb{Z}/8\mathbb{Z}$, $f(n) = [3n]$.
The equation $3n = 0$ modulo eight is equivalent to $n = 0$ modulo eight, so the kernel is $8\mathbb{Z}$. Every residue $[a]$ is $f(3a)$ because $9a = a$ modulo eight; the image is the whole target.
Check your argument. Give explicit reasons for both the kernel and surjectivity.

MM5020-U3.02 · E · 15 min
Learning objective: U3.1 Normality
Show the kernel of the determinant map $\mathrm{GL}_2(\mathbb{R}) \to \mathbb{R}^\times$ is normal.
$\det(AB) = \det(A)\det(B)$, so determinant is a group homomorphism. Its kernel is $\mathrm{SL}_2(\mathbb{R})$. Every homomorphism kernel is normal; alternatively $\det(ABA^{-1}) = \det(B) = 1$ for B in the kernel.
Check your argument. Identify the kernel and cite or prove the applicable normality criterion.

MM5020-U3.03 · E · 15 min
Learning objective: U3.2 Quotient size
Let $|G| = 60$ and $N \lhd G$ have order twelve. Determine $|G/N|$ and its group structure.
The quotient has $\frac{60}{12} = 5$ elements. Since five is prime, every nonidentity quotient element generates it, so $G/N \cong C_5$.
Check your argument. Use normality for the quotient group and Lagrange for its size.

MM5020-U3.04 · E · 15 min
Learning objective: U3.1 Homomorphisms
Show a homomorphism $f : G \to H$ satisfies $f(g^n) = f(g)^n$ for every integer n.
Induction and multiplication preservation prove the identity for positive n. For $n = 0$, both sides are the identity because $f(e) = e$. If $n = -m < 0$, then $f(g^n) = f\big((g^{-1})^m\big) = f(g^{-1})^m = f(g)^{-m}$.
Check your argument. Do not omit the negative or zero cases.

MM5020-U3.05 · C · 30 min
Learning objective: U3.2 First isomorphism theorem
Identify $\mathbb{R}^\times/\{1, -1\}$ using a natural homomorphism.
The map $x \mapsto |x|$ is a surjective multiplicative homomorphism onto $\mathbb{R}_{>0}$ with kernel $\{1, -1\}$. The first isomorphism theorem gives the quotient isomorphism $[x] \mapsto |x|$.
Check your argument. Specify the target operation and compute the kernel.

MM5020-U3.06 · C · 30 min
Learning objective: U3.1 Normality
Prove the center $Z(G)$ is a normal subgroup.
If a, b commute with every g, then so does ab^{-1}, since $gab^{-1} = agb^{-1} = ab^{-1}g$. The identity is central, so this is a subgroup. For $z \in Z(G)$, $gzg^{-1} = z$, proving normality.
Check your argument. Distinguish closure from the separate conjugation argument.

MM5020-U3.07 · C · 30 min
Learning objective: U3.2 Quotient maps
For positive integers m, n, when is $[a]_m \mapsto [a]_n$ well defined? When it is, find its kernel.
Necessity follows because $[m]_m = [0]_m$ forces $[m]_n = 0$. If $n|m$, a difference divisible by m is divisible by n, proving sufficiency. The kernel is the set of multiples of n modulo m, namely $\{[0], [n], \ldots, \big[\big(\frac{m}{n} - 1\big)n\big]\}$, of order $\frac{m}{n}$.
Check your argument. Prove necessity and sufficiency, then list the kernel in the correct source group.

MM5020-U3.08 · C · 30 min
Learning objective: U3.2 Correspondence
List all subgroups of $\mathbb{Z}/12\mathbb{Z}$ by using subgroups of $\mathbb{Z}$ containing $12\mathbb{Z}$.
The possible d are $1, 2, 3, 4, 6, 12$. Their images are $d\mathbb{Z}/12\mathbb{Z}$, generated by $[d]$, with orders $\frac{12}{d}$. The correspondence theorem proves there are no others.

Check your argument. Pay attention to the reversal between divisibility and ideal/subgroup containment.

MM5020-U3.09 · C · 30 min
Learning objective: U3.1 Cyclic quotient by center

Prove that if $G/Z(G)$ is cyclic, then G is abelian.

Choose $gZ(G)$ generating the quotient. Then write $x = g^r z$ and $y = g^s w$ for integers r, s and $z, w \in Z(G)$, since their cosets are powers of the generating coset. Since z, w commute with everything, $xy = g^{r+s}zw = g^{s+r}wz = yx$. Hence G is abelian.

Check your argument. Explain why the expressions cover all elements and why the central factors may move.

MM5020-U3.10 · A · 50 min
Learning objective: U3.2 Universal property

Prove that a homomorphism $f : G \to H$ factors through G/N exactly when $N \subseteq \ker f$, and prove uniqueness of the factor.

If $f = \overline{f}\pi$, then $n \in N$ maps to $\overline{f}(N) = e$, so $N \subseteq \ker f$. Conversely the containment makes equal cosets have equal images, because $b^{-1}a \in N$ implies $f(a) = f(b)$. The candidate respects products by the homomorphism law. Every coset is $\pi(g)$, forcing any factor's value there to be $f(g)$, so it is unique.

Check your argument. Give both directions and all well-definedness details.

MM5020-U3.11 · A · 50 min
Learning objective: U3.2 Abelianization

Let $[G, G]$ be generated by all commutators $aba^{-1}b^{-1}$. Show it is normal, its quotient is abelian, and every map from G to an abelian group factors uniquely through this quotient.

Conjugating a commutator by g gives the commutator of gag^{-1} and gbg^{-1}, so conjugation preserves the generated subgroup. In the quotient each commutator is the identity, precisely the equation making the cosets of a, b commute. If f has abelian target, the image of every commutator is the identity, so $[G, G] \subseteq \ker f$. The quotient universal property gives the unique factorization.

Check your argument. Show normality for the generated subgroup, not only for individual generators.

MM5020-U3.12 · A · 50 min
Learning objective: U3.1 Failure of converse to Lagrange

Prove that A_4 has no subgroup of order six. You may use that three-cycles generate A_4.

Decisive idea. Turn the proposed subgroup into a quotient, then show that every generator would disappear in that quotient.

Finding the approach. An order-six subgroup would have index two. That unusually small index is more useful than listing six-element subsets: it gives a quotient with only two elements.

Argument. Such a subgroup would be normal by the index-two theorem, yielding a surjective homomorphism $A_4 \to C_2$. The image of an element of order three has order dividing both three and two, hence is the identity. Since three-cycles generate A_4, the homomorphism would be trivial, contradicting surjectivity.

For a three-cycle g, its image h satisfies $h^3 = e$. In the order-two target it also satisfies $h^2 = e$, so $h = h^3(h^2)^{-1} = e$. A homomorphism that kills every generator kills every product of generators. Yet the quotient map must hit the nonidentity coset, which is the contradiction.

A tempting dead end. Lagrange only says a subgroup order must divide twelve. It does not construct a subgroup for every divisor. Counting subsets of six elements ignores closure and normality.

Hypotheses. Use the given generation by three-cycles and the index-two normality theorem. The argument depends on the coprime orders three and two, not merely on the quotient being small.

Check your argument. Explain why the image order divides the original order and why generators determine the map.

MM5020-U3.13 · A · 60 min · Optional challenge
Learning objective: U3 Transfer: A large cyclic kernel

Let $G = (\mathbb{Z}/12\mathbb{Z}) \times (\mathbb{Z}/18\mathbb{Z})$. Define $f : G \to \mathbb{Z}/6\mathbb{Z}$ by $f(a, b) = a + b$. Determine the kernel and prove that it is cyclic; give a generator and the quotient up to isomorphism.

Decisive idea. Find one compatible motion of both coordinates, then certify that it exhausts the kernel by its order.

Finding the approach. The relation defining the kernel suggests moving one coordinate up while moving the other down. Its order can be compared with the kernel's size.
Argument. The multiples of twelve and eighteen vanish modulo six, so f is well defined; addition makes it a homomorphism. It is onto since $f(a, 0)$ reaches every residue modulo six. Hence its kernel has $12 \cdot \frac{18}{6} = 36$ elements. The element $g = (1, -1)$ is in the kernel. A positive integer n kills g exactly when both twelve and eighteen divide n, whose least possibility is 36. Thus g generates a subgroup of order 36 already contained in the kernel, so it is the whole kernel. The first isomorphism theorem gives $G/\ker(f) \cong \mathbb{Z}/6\mathbb{Z}$.
A tempting dead end. Listing possible coordinate pairs can find elements but does not by itself establish cyclicity. Nor does the fact that a kernel has order thirty-six force it to be cyclic.
Hypotheses. Both source moduli must be divisible by six for the stated formula to descend to residue classes. Cardinality arguments use finite groups.
Check your argument. Prove well-definedness, surjectivity, the generator's exact order, and the quotient identification.

4 Unit 4 · Actions and simple groups

MM5020-U4.01 · E · 15 min
Learning objective: U4.1 Actions
Verify that S_n acts on $\{1, \ldots, n\}$ by evaluation.
The identity permutation fixes every element. Also $(\sigma\tau) \cdot i = \sigma(\tau(i)) = \sigma\tau \cdot i$ by the definition of composition. These are exactly the action axioms.
Check your argument. Explicitly check both axioms.

MM5020-U4.02 · E · 15 min
Learning objective: U4.1 Stabilizers
Find the stabilizer of one in the natural S_4 action and its order.
The stabilizer contains exactly the permutations fixing one. Restriction to $\{2, 3, 4\}$ identifies it with S_3, so it has six elements. Orbit–stabilizer agrees: $\frac{24}{6} = 4$.
Check your argument. Give the subgroup description as well as its size.

MM5020-U4.03 · E · 15 min
Learning objective: U4.2 Conjugacy
Compute $(123)(12)(123)^{-1}$.
The result is (23), by the formula $\tau(ab)\tau^{-1} = (\tau(a)\tau(b))$.
Check your argument. Identify the conjugating permutation and its two relevant images.

MM5020-U4.04 · E · 15 min
Learning objective: U4.2 Simplicity
Which nontrivial finite cyclic groups are simple?
Exactly the cyclic groups of prime order. Prime order permits no proper nontrivial subgroup by Lagrange. For composite order $n = ab$ with $a, b > 1$, the subgroup generated by g^a has order b, hence is proper, nontrivial, and normal.
Check your argument. Address both prime and composite orders.

MM5020-U4.05 · C · 30 min
Learning objective: U4.1 Kernel of an action
Show the kernel of the homomorphism associated with an action is the intersection of all stabilizers.
An element g is in the homomorphism kernel exactly when its associated permutation is the identity, meaning $g \cdot x = x$ for every x. This is equivalent to membership in every G_x, so the kernel equals $\cap_{x \in X} G_x$. It is normal because it is a kernel.
Check your argument. Explain the equivalence point by point.

MM5020-U4.06 · C · 30 min
Learning objective: U4.1 Conjugate stabilizers
Prove $G_{h \cdot x} = hG_xh^{-1}$.
$gh \cdot x = h \cdot x$ holds exactly when $(h^{-1}gh) \cdot x = x$. Thus g stabilizes $h \cdot x$ exactly when $h^{-1}gh \in G_x$, which is equivalent to $g \in hG_xh^{-1}$.
Check your argument. Prove equality through an if-and-only-if membership calculation.

MM5020-U4.07 · C · 30 min
Learning objective: U4.2 Class equation
Use conjugacy classes to show the center of S_3 is trivial.
The identity is a singleton class. The three transpositions form one class and the two three-cycles another, by cycle-type conjugacy. Only elements in singleton conjugacy classes are central, so $Z(S_3) = \{e\}$.
Check your argument. Explain why a singleton conjugacy class characterizes a central element.

MM5020-U4.08 · C · 30 min
Learning objective: U4.2 Prime-power groups
Prove a group of order p^2 is abelian.
The class equation gives a nontrivial center, whose order is p or p^2 by Lagrange. In the latter case the group is abelian. In the former, $G/Z(G)$ has prime order and is cyclic; U3.09 then makes G abelian,

forcing the center to be all of G and contradicting its supposed order p.
Check your argument. Treat both possible center sizes and justify the cyclic-quotient step.

MM5020-U4.09 · C · 30 min
Learning objective: U4.1 Orbit counting
Count two-colorings of four labeled square vertices up to rotation, with reflections excluded.
The identity fixes all $2^4 = 16$. A quarter or three-quarter rotation forces all vertices to share one color, giving two each. A half-turn permits independent colors on two opposite pairs, giving four. The orbit count is $\frac{16+2+4+2}{4} = 6$.
Check your argument. Use precisely the rotation group requested and explain each fixed-point count.

MM5020-U4.10 · A · 50 min
Learning objective: U4.2 Simple prime-power groups
Classify finite simple groups whose order is a power of a prime.
For order p^n with $n \geq 1$, the center is nontrivial by the class equation. It is normal, so simplicity forces $Z(G) = G$. Thus G is abelian. A nontrivial finite abelian simple group is cyclic of prime order, so here the only possibility is C_p. Conversely C_p is simple.
Check your argument. Prove the converse and exclude all higher prime powers.

MM5020-U4.11 · A · 50 min
Learning objective: U4.1 Coset actions
Let H have finite index n in G. Show there is a normal subgroup $N \subseteq H$ with $[G : N] \leq n!$.
Left multiplication defines an action on the n cosets. Let N be its kernel; it is normal. An element of N fixes the coset H, so it belongs to H. The first isomorphism theorem identifies G/N with the image subgroup of S_n, so its order is at most $n!$. This also proves the quotient is finite even if G is infinite.
Check your argument. Do not assume H itself is normal or G finite.

MM5020-U4.12 · A · 50 min
Learning objective: U4.2 Faithful actions of simple groups
Prove every nontrivial action of a simple group is faithful. Deduce that A_5 cannot act nontrivially on a set of four elements.
Decisive idea. Use the kernel to convert an action into either no information at all or an embedding.
Finding the approach. An action is a homomorphism to a symmetric group. Simplicity restricts its kernel before any individual orbit needs to be computed.
Argument. The action kernel is either trivial or the whole simple group. A nontrivial action rules out the latter, so it is faithful. Since A_5 is simple of order sixty, a nontrivial four-point action would give an injection into S_4, which has only twenty-four elements, impossible.
The kernel consists of elements fixing every point, hence is a normal subgroup. If it were the whole group, every group element would act as the identity permutation, exactly the trivial action. Otherwise simplicity leaves the identity subgroup. An injective map between finite groups is an injection of their underlying sets; sixty distinct elements cannot fit among twenty-four permutations.
A tempting dead end. Nontrivial does not mean transitive, and faithful does not mean that each point stabilizer is trivial. The relevant subgroup is the intersection of all stabilizers, not one stabilizer.
Hypotheses. The group must be simple and the action must be nontrivial. For the final obstruction use the previously proved simplicity and order of A_5.
Check your argument. Here "nontrivial action" means at least one group element moves at least one point.

MM5020-U4.13 · A · 60 min · Optional challenge
Learning objective: U4 Transfer: Four-position ornaments
An ornament has four positions at the vertices of a square, each colored with one of three colors. Two colorings are identified if a rotation or reflection of the square relates them. How many ornaments are there? Explain why simply dividing the number of colorings by eight fails.
Decisive idea. Count fixed objects by cycles, then average to account for unequal orbit sizes.
Finding the approach. The identification is by an action, and fixed colorings are much easier to count than equivalence classes directly. Count cycles rather than drawing every ornament.
Argument. The identity has four cycles and fixes $3^4 = 81$ colorings. The two quarter turns each have one cycle and fix three; the half turn has two cycles and fixes nine. Each of two diagonal reflections fixes

two vertices and swaps the other two, so has three cycles and fixes 27. Each of two edge-axis reflections has two swaps and fixes nine. The fixed-point average is $\frac{81+2\cdot3+9+2\cdot27+2\cdot9}{8} = 21$. Division of 81 by eight fails because constant colorings have singleton orbits while many other colorings have larger orbits.

A tempting dead end. Treating all four reflections alike loses their different cycle structures on vertices. Using only rotations also identifies fewer pairs than the question permits.

Hypotheses. The symmetry group has order eight and acts on vertex colorings. If positions were edges, the two reflection types would exchange roles; if reflections were disallowed, the answer would change.

Check your argument. List all five symmetry types and explain the number of fixed colorings for each.

5 Unit 5 · Rings and their quotients

MM5020-U5.01 · E · 15 min
Learning objective: U5.1 Units
Find every unit in $\mathbb{Z}/10\mathbb{Z}$ and its inverse.
The units are $[1], [3], [7], [9]$. Their inverses are respectively $[1], [7], [3], [9]$, since $3 \cdot 7 = 21 = 1$ modulo ten and $9^2 = 81 = 1$. The remaining residues share a factor with ten and cannot have a product congruent to one.
Check your argument. Show why the list is exhaustive.

MM5020-U5.02 · E · 15 min
Learning objective: U5.1 Zero divisors
Give nonzero matrices $A, B \in M_2(\mathbb{Q})$ with $AB = 0$.
Take $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. Both are nonzero and their product is the zero matrix.
Check your argument. Verify the product explicitly.

MM5020-U5.03 · E · 15 min
Learning objective: U5.2 Ideals
Show $6\mathbb{Z}$ is an ideal of $\mathbb{Z}$, and describe the identity of its quotient.
Differences of multiples of six are multiples of six, and $r(6k) = 6(rk)$, so it is an ideal. The quotient identity is $[1] = 1 + 6\mathbb{Z}$, since $[1][a] = [a]$.
Check your argument. Do not confuse the zero coset with the multiplicative identity.

MM5020-U5.04 · E · 15 min
Learning objective: U5.1 Quaternions
Find the inverse of $q = 1 + 2i - j$.
The conjugate is $1 - 2i + j$ and the norm is $1^2 + 2^2 + (-1)^2 = 6$. Thus $q^{-1} = \frac{1-2i+j}{6}$. Multiplication in either order gives one by the norm identity.
Check your argument. Use the squared norm rather than its square root as denominator.

MM5020-U5.05 · C · 30 min
Learning objective: U5.2 Quotient criteria
Determine whether (5) and (6) are prime or maximal in $\mathbb{Z}$.
$\mathbb{Z}/5\mathbb{Z}$ is a field, so (5) is maximal and prime. Modulo six, $[2][3] = 0$ with both factors nonzero, so (6) is not prime and therefore not maximal.
Check your argument. Justify both the positive and negative conclusions using quotient properties.

MM5020-U5.06 · C · 30 min
Learning objective: U5.2 Evaluation
Show $\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$.
Evaluation at i is a unital ring homomorphism and is surjective because $a + bi$ is the image of $a + bx$. Dividing a polynomial by $x^2 + 1$ leaves $a + bx$; its value at i vanishes only if $a = b = 0$. Thus the kernel is exactly $(x^2 + 1)$, and the first isomorphism theorem applies.
Check your argument. Prove the kernel equality, not just one containment.

MM5020-U5.07 · C · 30 min
Learning objective: U5.1 Characteristic
Prove a nonzero domain has characteristic zero or a prime.
If the characteristic is a positive composite $n = ab$ with $1 < a, b < n$, then $(a \cdot 1)(b \cdot 1) = n \cdot 1 = 0$. Minimality of n makes both factors nonzero, contradicting the domain property. Thus any positive characteristic is prime.
Check your argument. Use minimality to establish that both factors are nonzero.

MM5020-U5.08 · C · 30 min
Learning objective: U5.2 Ideals and units
Prove an ideal contains a unit if and only if it is the whole ring. Deduce that a field has exactly two ideals.
If $u \in I$ is a unit, absorption gives $u^{-1}u = 1 \in I$, then $r = r \cdot 1 \in I$ for every r. Conversely the whole ring contains one, a unit. In a field, every nonzero ideal contains a nonzero element, hence a unit, so the

only ideals are zero and the whole field.
Check your argument. Explain both implications and specify nonzero ideals in the deduction.

MM5020-U5.09 · C · 30 min
Learning objective: U5.2 Noncommutative ideals
Show the matrices with zero second column form a left ideal of $M_2(K)$ but not a two-sided ideal.
The set is an additive subgroup and left multiplication preserves the zero second column. But E_{11} lies in it, while $E_{11}E_{12} = E_{12}$ does not. Hence it is not closed under right multiplication and is not two-sided.
Check your argument. Check the specified side of multiplication carefully.

MM5020-U5.10 · A · 50 min
Learning objective: U5.2 Chinese remainder theorem
Prove $\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ for coprime positive m, n.
Let $f(a) = ([a]_m, [a]_n)$. Its kernel consists of integers divisible by both m, n, hence by mn because they are coprime. Choose $um + vn = 1$. For a target $([r]_m, [s]_n)$, the integer $rvn + sum$ maps to it. Thus f is surjective with the stated kernel, and the ring isomorphism theorem proves the claim.
Check your argument. Verify both congruences in the surjectivity construction.

MM5020-U5.11 · A · 50 min
Learning objective: U5.2 Matrix ring ideals
Prove every two-sided ideal of $M_{n(K)}$, for a field K, is zero or the whole ring.
If $A \in I$ has nonzero entry a_{ij}, then $E_{ri}AE_{js} = a_{ij}E_{rs} \in I$ for every r, s. Multiplying by the scalar inverse gives every matrix unit in I. Their linear combinations yield every matrix, so $I = M_{n(K)}$. Thus any nonzero ideal is the whole ring.
Check your argument. Use two-sided absorption and the invertibility of the nonzero coefficient.

MM5020-U5.12 · A · 50 min
Learning objective: U5.1 Division rings
Prove quaternion conjugation reverses products and deduce multiplicativity of $N(q) = q\overline{q}$.
Decisive idea. Conjugation reverses multiplication; the norm becomes multiplicative because the middle conjugate pair is central.
Finding the approach. Quaternion multiplication is not commutative. Check what conjugation does to a small basis, then use bilinearity rather than expanding two arbitrary four-term expressions without a plan.
Argument. For the basis $1, i, j, k$, the rule follows from the multiplication table: conjugation negates each imaginary basis element, and reversing two distinct imaginary factors negates their product. Equal factors give the same real value in either order. Real bilinearity extends the identity to all q, r. Hence $N(qr) = qr\overline{rq} = qN(r)\overline{q} = N(r)N(q)$, since $N(r)$ is real and central.
For distinct imaginary basis elements u, v, write $uv = w$ or $-w$ with w imaginary. Then $vu = -uv$ and $\overline{uv} = -uv = \overline{vu}$. For $u = v$, both sides are -1; factors equal to 1 are immediate. Distributing over real coefficients proves the identity for arbitrary quaternions. Finally $q\overline{q}$ is the real sum of four coordinate squares, which justifies moving $N(r)$ through q.
A tempting dead end. Writing $\overline{qr} = \overline{qr}$ keeps the wrong order. Taking $q = i, r = j$ gives $-k$ on the left and k on that incorrect right side.
Hypotheses. Real coefficients commute with the basis elements, and the norm is a central real scalar. Neither fact licenses commuting arbitrary quaternions.
Check your argument. Check equal as well as distinct imaginary factors, and justify the central scalar step.

MM5020-U5.13 · A · 60 min · Optional challenge
Learning objective: U5 Transfer: Coordinates hidden in residues
Find all e in $\mathbb{Z}/12\mathbb{Z}$ satisfying $e^2 = e$. Construct an explicit ring isomorphism from $\mathbb{Z}/12\mathbb{Z}$ to $(\mathbb{Z}/3\mathbb{Z}) \times (\mathbb{Z}/4\mathbb{Z})$ and use it to explain your list.
Decisive idea. Separate residue coordinates to turn one equation with zero divisors into two tiny equations.
Finding the approach. The requested product suggests recording two residues. Constructing its inverse will also reveal how to combine idempotents in the two smaller rings.

Argument. Send x to $(x \bmod 3, x \bmod 4)$. This is a well-defined ring homomorphism. Given (a, b), the class $4a + 9b$ is independent of representatives, since changes by three and four alter it by multiples of twelve. Its residues are (a, b). Conversely $4x + 9x = 13x = x \bmod 12$, so these maps are inverse. In $\mathbb{Z}/3\mathbb{Z}$, the idempotents are zero and one since it is a field. Directly testing zero, one, two, three modulo four gives only zero and one there too. The four coordinate choices therefore give $0, 4, 9, 1$ modulo twelve.
A tempting dead end. Solving $e(e-1) = 0$ by declaring a factor zero is invalid in a ring with zero divisors.
Hypotheses. The two moduli are coprime, as reflected by the explicit coefficients four and nine. The proof gives an inverse, not just a surjection between unspecified rings.
Check your argument. Verify both composites and explain why the zero-product inference fails modulo twelve.

6 Unit 6 · Polynomials and factorization

MM5020-U6.01 · E · 15 min
Learning objective: U6.1 Division
Divide $x^3 + 2x + 1$ by $x^2 + 1$ over $\mathbb{Q}$.
The quotient is x and remainder $x + 1$, since $x^3 + 2x + 1 = x(x^2 + 1) + (x + 1)$ and the remainder has degree less than two.
Check your argument. Give and verify the division identity and degree bound.

MM5020-U6.02 · E · 15 min
Learning objective: U6.1 Bézout
Find integers u, v with $30u + 21v = 3$.
The identity is $3 = -2 \cdot 30 + 3 \cdot 21$, so $u = -2, v = 3$. The Euclidean divisions $30 = 21 + 9$ and $21 = 2 \cdot 9 + 3$ explain the back substitution.
Check your argument. Provide the identity, not just the gcd.

MM5020-U6.03 · E · 15 min
Learning objective: U6.2 Irreducibility
Prove $x^3 - 5$ is irreducible over $\mathbb{Q}$.
The polynomial is primitive and its leading coefficient is not divisible by five. Every lower coefficient is divisible by five, while the constant term is not divisible by twenty-five. Eisenstein therefore proves irreducibility over $\mathbb{Q}$.
Check your argument. Check every hypothesis, including the square condition.

MM5020-U6.04 · E · 15 min
Learning objective: U6.1 Roots
Factor $x^2 + 1$ over $\mathbb{F}_5$.
The roots are two and three, so $x^2 + 1 = (x - 2)(x - 3)$ over $\mathbb{F}_5$. Expanding gives $x^2 - 5x + 6 = x^2 + 1$ modulo five.
Check your argument. State the coefficient field in the factorization.

MM5020-U6.05 · C · 30 min
Learning objective: U6.1 Modular inverses
Find the inverse of $x + 1$ in $\mathbb{Q}[x]/(x^2 + 1)$.
Modulo $x^2 + 1$, $(x + 1)(x - 1) = -2$. Therefore the inverse is $\frac{1-x}{2}$. Direct multiplication gives $\frac{1-x^2}{2} = 1$ in the quotient.
Check your argument. Reduce using the quotient relation and check the final product.

MM5020-U6.06 · C · 30 min
Learning objective: U6.2 Reduction test
Prove $x^3 + x + 1$ is irreducible over $\mathbb{Q}$ by reducing modulo two.
The reduction has value one at both zero and one, so has no root in $\mathbb{F}_2$ and is irreducible there. The integer polynomial is primitive and monic, so its degree is preserved on reduction. Gauss's lemma and the reduction criterion prove irreducibility over $\mathbb{Q}$.
Check your argument. Explain why the root test suffices for degree three and why reduction applies.

MM5020-U6.07 · C · 30 min
Learning objective: U6.2 Failure outside PIDs
Show 2 is irreducible but not prime in $\mathbb{Z}\left[\sqrt{-5}\right]$.
Define $N\left(a + b\sqrt{-5}\right) = a^2 + 5b^2$. Multiplying an element by its conjugate gives this norm; conjugation preserves products in this commutative ring, so the norm is multiplicative. Its value is one exactly on the units $1, -1$. If $2 = uv$, multiplicativity of the norm gives $4 = N(u)N(v)$. The only possible nonunit factorization would require both norms to be two, impossible from $a^2 + 5b^2 = 2$. Thus two is irreducible. It divides six, but neither $1 + \sqrt{-5}$ nor $1 - \sqrt{-5}$ is divisible by two, because their coefficients are not both even. Hence two is not prime.
Check your argument. Explain why norm one means a unit and why divisibility by two forces even coefficients.

MM5020-U6.08 · C · 30 min
Learning objective: U6.1 Principal ideals

Show $K[x]/(f)$ is a field if and only if f is irreducible, assuming f is nonconstant.

If f is irreducible, a nonzero class $[g]$ has $\gcd(f,g)=1$, so a Bézout identity gives its inverse. Conversely, if $f=ab$ with both factors of positive smaller degree, then $[a],[b]$ are nonzero classes whose product is zero, impossible in a field.

Check your argument. Prove the factors give nonzero classes by their degrees.

MM5020-U6.09 · C · 30 min
Learning objective: U6.2 Primitive products

Show the product of two primitive integer polynomials is primitive.

For any prime p, primitiveness means neither factor becomes the zero polynomial modulo p. Their product is therefore nonzero in the domain $\mathbb{F}_{p[x]}$. Thus no prime divides every coefficient of the integer product, which is exactly primitiveness.

Check your argument. The argument must apply to every prime.

MM5020-U6.10 · A · 50 min
Learning objective: U6.2 Shifted Eisenstein

For prime p, prove $1+x+\ldots+x^{p-1}$ is irreducible over $\mathbb{Q}$.

Substitute $x=y+1$. The polynomial becomes $\frac{(y+1)^p-1}{y}$, a polynomial with coefficients $\binom{p}{k}$ for the powers y^{k-1}, $1\le k\le p$. Its leading coefficient is one, all others are divisible by p, and its constant term is p, not divisible by p^2. Eisenstein proves it irreducible. Substitution $x\mapsto y+1$ is an invertible ring change of variable, with inverse $y\mapsto x-1$, so it preserves reducibility and the original polynomial is irreducible.

Check your argument. Justify both binomial divisibility and invariance under the change of variable.

MM5020-U6.11 · A · 50 min
Learning objective: U6.1 Polynomial Chinese remainder

If $f,g\in K[x]$ are coprime, prove $K[x]/(fg)\cong K[x]/(f)\times K[x]/(g)$.

The map $h\mapsto\left([h]_f,[h]_g\right)$ is a unital homomorphism. A polynomial divisible by both coprime f,g is divisible by fg: from $h=fk$ and $g|h$, primeness of irreducible factors of g gives $g|k$, or multiply a Bézout identity by k. Its kernel is therefore (fg). Choose u,v with $uf+vg=1$. Given a,b, the polynomial $avg+buf$ has the prescribed remainders. Surjectivity and the first isomorphism theorem finish.

Check your argument. Compute the kernel and explicitly verify both residues.

MM5020-U6.12 · A · 50 min
Learning objective: U6.2 Nonprincipal ideals

Prove $\mathbb{Z}[x]$ is a UFD but not a PID.

Decisive idea. A common divisor controls a possible principal generator; a quotient map then proves that no such generator can work.

Finding the approach. To separate UFD from PID, use different evidence for the two properties: a factorization theorem for the first and one carefully chosen ideal for the second.

Argument. Since $\mathbb{Z}$ is a UFD, Gauss's theorem makes $\mathbb{Z}[x]$ a UFD. If $(2,x)=(d)$, then $d|2$, so d is a constant integer. Also $d|x$, forcing $d=1$ or -1. That would make the ideal the whole ring. But evaluation at zero followed by reduction modulo two is a homomorphism killing both generators and not one, so the ideal is proper. Contradiction.

If a nonzero polynomial d divides the constant 2, degree additivity in the domain $\mathbb{Z}[x]$ forces degree zero. A constant integer dividing the polynomial x divides its coefficient 1, so it is a unit. The evaluation map $f\mapsto f(0)\bmod 2$ sends every element $2a(x)+xb(x)$ to zero but sends 1 to one; thus the ideal cannot equal the whole ring.

A tempting dead end. Proving that 2 and x have no common nonunit divisor does not prove that their ideal is the unit ideal. That implication holds in a Bézout domain, which is precisely more than has been established here.

Hypotheses. Gauss's theorem requires the coefficient ring to be a UFD. Degree additivity uses the absence of zero divisors. The ideal argument distinguishes divisibility from linear combination.

Check your argument. Separate the UFD proof from the argument excluding a principal generator.

MM5020-U6.13 · A · 60 min · Optional challenge

Learning objective: U6 Transfer: Two quartics without rational roots

Determine which of x^4+1 and x^4+4 are irreducible over $\mathbb{Q}$. Give complete factorizations into irreducibles in $\mathbb{Z}[x]$ and explain why testing rational roots does not settle either question.

Decisive idea. Change coordinates to expose a prime-divisibility obstruction, and distinguish it from a constructive factorization.

Finding the approach. Degree four permits a factorization into quadratics without any rational root. Look for a transformation that proves impossibility in one case and an identity that exhibits it in the other.

Argument. The translated first polynomial is $x^4+4x^3+6x^2+4x+2$, which is Eisenstein at two: all nonleading coefficients are divisible by two and the constant is not divisible by four. It is irreducible over $\mathbb{Q}$. Substitution $x \mapsto x+1$ is an automorphism of $\mathbb{Q}[x]$, with inverse $x \mapsto x-1$, so x^4+1 is irreducible too. Meanwhile $x^4+4 = (x^2+2)^2-(2x)^2 = (x^2-2x+2)(x^2+2x+2)$. Both quadratics have discriminant -4, hence no rational root and are irreducible over $\mathbb{Q}$. All these polynomials are primitive; Gauss's lemma transfers the assertions to $\mathbb{Z}[x]$, up to units.

A tempting dead end. The rational-root test only excludes linear factors. It misses the displayed pair of quadratic factors of the second polynomial.

Hypotheses. Translation must be invertible to preserve reducibility in both directions. Gauss's lemma applies to primitive integer polynomials; the coefficient content must not be ignored.

Check your argument. Explain the inverse translation, every Eisenstein condition, and irreducibility of both quadratic factors.

7 Unit 7 · Fields as vector spaces

MM5020-U7.01 · E · 15 min
Learning objective: U7.1 Fractions
Prove $\frac{a}{b} = 0$ in the fraction field of a domain exactly when $a = 0$.
Equality with $\frac{0}{1}$ means $a \cdot 1 = b \cdot 0$, hence $a = 0$. Conversely a zero numerator gives that equality directly.
Check your argument. Use the equivalence relation defining fractions.

MM5020-U7.02 · E · 15 min
Learning objective: U7.2 Minimal polynomials
Find the minimal polynomial of $\sqrt{7}$ over $\mathbb{Q}$ and a basis of its generated field.
The minimal polynomial is $x^2 - 7$, since it is monic, vanishes at $\sqrt{7}$, and is irreducible. Hence the extension has degree two and basis $1, \sqrt{7}$.
Check your argument. Verify irreducibility rather than merely exhibit a vanishing polynomial.

MM5020-U7.03 · E · 15 min
Learning objective: U7.1 Tower law
If $[L : K] = 3$ and $[M : L] = 4$, find $[M : K]$. Can an intermediate field between K and M have degree five over K?
The total degree is twelve. If $K \subseteq F \subseteq M$ had $[F : K] = 5$, the tower law would give $12 = [M : F] \cdot 5$, impossible.
Check your argument. State the tower in each application.

MM5020-U7.04 · E · 15 min
Learning objective: U7.2 Power-basis arithmetic
With $\alpha^3 = 2$, simplify $\alpha^5 + \alpha^4$ in the power basis.
$\alpha^5 = 2\alpha^2$ and $\alpha^4 = 2\alpha$, so the result is $2\alpha^2 + 2\alpha$.
Check your argument. Keep coefficients in the base field and powers below the minimal-polynomial degree.

MM5020-U7.05 · C · 30 min
Learning objective: U7.2 Inverses
Find $\left(2 + \sqrt{3}\right)^{-1}$ in $\mathbb{Q}\left(\sqrt{3}\right)$ and explain why every nonzero $a + b\sqrt{3}$ has an inverse in this field.
The requested inverse is $2 - \sqrt{3}$ since the product is one. In general the inverse is $\left(a - b\sqrt{3}\right)/(a^2 - 3b^2)$. If the denominator were zero with a nonzero numerator, then $b \neq 0$ and $\left(\frac{a}{b}\right)^2 = 3$, contradicting irrationality of $\sqrt{3}$. Thus the denominator is nonzero.
Check your argument. Check the denominator cannot vanish for a nonzero element.

MM5020-U7.06 · C · 30 min
Learning objective: U7.2 Algebraicity
Prove every element of a finite extension is algebraic.
Put $d = [L : K]$. For any $\alpha \in L$, the $d + 1$ powers $1, \alpha, \ldots, \alpha^d$ are linearly dependent. This supplies coefficients $c_i \in K$, not all zero, with $\sum_{i=0}^{d} c_i \alpha^i = 0$. The corresponding nonzero polynomial annihilates α, so α is algebraic.
Check your argument. Explain why the polynomial is nonzero.

MM5020-U7.07 · C · 30 min
Learning objective: U7.2 Change of generator
Find the minimal polynomial of $1 + \sqrt{2}$ over $\mathbb{Q}$.
If $\alpha = 1 + \sqrt{2}$, then $(\alpha - 1)^2 = 2$, so $\alpha^2 - 2\alpha - 1 = 0$. The polynomial $x^2 - 2x - 1$ has roots $1 \pm \sqrt{2}$, neither rational, hence is irreducible and is the minimal polynomial.
Check your argument. Show irreducibility and monicity.

MM5020-U7.08 · C · 30 min
Learning objective: U7.1 Fraction universal property
Explain why the reduction map $\mathbb{Z} \to \mathbb{F}_5$ cannot extend to a unital homomorphism $\mathbb{Q} \to \mathbb{F}_5$.
Such an extension would send five to zero. Applying it to $5\left(\frac{1}{5}\right) = 1$ yields $0 \cdot f\left(\frac{1}{5}\right) = 1$, a contradiction in the nonzero field $\mathbb{F}_5$. This shows why the fraction-field extension theorem requires nonzero denominators

to have nonzero images.
Check your argument. Identify the exact relation whose image is impossible.

MM5020-U7.09 · C · 30 min
Learning objective: U7.2 Root adjunction
Construct a field containing a root of $x^3 + x + 1$ over $\mathbb{F}_2$ and list a basis over $\mathbb{F}_2$.
At zero and one the polynomial has value one, so as a cubic it is irreducible. Therefore $K = \mathbb{F}_2[x]/(x^3 + x + 1)$ is a field. Writing $\alpha = [x]$, the elements $1, \alpha, \alpha^2$ form a basis and $\alpha^3 = \alpha + 1$.
Check your argument. Explain why the quotient is a field and why the powers are a basis.

MM5020-U7.10 · A · 50 min
Learning objective: U7.1 Compositum degree
Prove $\left[\mathbb{Q}\left(\sqrt{2}, \sqrt{5}\right) : \mathbb{Q}\right] = 4$ and find a basis.
If $\sqrt{5} = a + b\sqrt{2}$ with rational a, b, squaring gives $5 = a^2 + 2b^2 + 2ab\sqrt{2}$, so $ab = 0$. If $b = 0$, then $a^2 = 5$; if $a = 0$, then $b^2 = \frac{5}{2}$. Neither is a rational square, as prime exponents in a rational square are even. Thus adjoining $\sqrt{5}$ has degree two over $\mathbb{Q}\left(\sqrt{2}\right)$, and the tower law gives four. Products of the two quadratic bases yield $1, \sqrt{2}, \sqrt{5}, \sqrt{10}$.
Check your argument. Exclude both cases and invoke the basis form of the tower law.

MM5020-U7.11 · A · 50 min
Learning objective: U7.2 Transitivity of algebraicity
Let L/K be algebraic and let α be algebraic over L. Prove α is algebraic over K, even when L/K is infinite.
Let $c_0, \ldots, c_d \in L$ be the coefficients of a nonzero polynomial annihilating α. The field $F = K(c_0, \ldots, c_d)$ is finite over K, since it is generated by finitely many algebraic elements. The element α is algebraic over F, so $F(\alpha)/F$ is finite. The tower law makes $F(\alpha)/K$ finite. Every element of a finite extension is algebraic, so α is algebraic over K.
Check your argument. Do not apply a finite-degree tower law directly to the potentially infinite extension L/K.

MM5020-U7.12 · A · 50 min
Learning objective: U7.2 A primitive generator
Show $\mathbb{Q}\left(\sqrt{2} + \sqrt{3}\right) = \mathbb{Q}\left(\sqrt{2}, \sqrt{3}\right)$ and find the minimal polynomial of the sum.
Decisive idea. Recover the separate generators from the sum and its inverse, then use the tower degree to certify minimality.
Finding the approach. Look for a second expression in the same field that separates the two radicals. Multiplying the sum by the difference is useful because the mixed terms cancel.
Argument. Put $\alpha = \sqrt{2} + \sqrt{3}$. Then $\alpha^{-1} = \sqrt{3} - \sqrt{2}$, so $\sqrt{3} = \frac{\alpha + \alpha^{-1}}{2}$ and $\sqrt{2} = \frac{\alpha - \alpha^{-1}}{2}$. This proves equality of the fields. Squaring twice gives $\left(\alpha^2 - 5\right)^2 = 24$, hence $\alpha^4 - 10\alpha^2 + 1 = 0$. The field has degree four by Lecture 13, so the minimal polynomial has degree four and must be this monic quartic. The product $\left(\sqrt{2} + \sqrt{3}\right)\left(\sqrt{3} - \sqrt{2}\right) = 1$ shows that the inverse is already available in $\mathbb{Q}(\alpha)$. For completeness, $\sqrt{3}$ is not in $\mathbb{Q}\left(\sqrt{2}\right)$: writing it as $a + b\sqrt{2}$ and squaring gives $ab = 0$, then either $a^2 = 3$ or $b^2 = \frac{3}{2}$, impossible for rational a, b. The tower therefore has degree four. The quartic that vanishes at α is monic of that degree, so divisibility by the minimal polynomial forces equality.
A tempting dead end. An equation of degree four satisfied by α supplies only an upper bound for its degree. Irreducibility or an independent degree computation is still necessary.
Hypotheses. These calculations take place in a field of characteristic zero, so inverses of the nonzero sum and of two exist. The degree argument uses genuine nonmembership of one radical in the first quadratic field.
Check your argument. Use the degree to prove minimality; a vanishing quartic alone is insufficient.

MM5020-U7.13 · A · 60 min · Optional challenge
Learning objective: U7 Transfer: A field containing three roots
Let a be the positive real cube root of two and let ω be a nonreal root of $x^2 + x + 1$. Determine the smallest subfield of $\mathbb{C}$ containing every root of $x^3 - 2$, and its degree over $\mathbb{Q}$.
Decisive idea. Prove minimality by recovering the proposed generators from the required roots, then compute the two tower steps separately.

Finding the approach. The roots tell us which generators are sufficient; their ratios tell us which are necessary. Then use the tower to determine degree instead of trying to write a six-degree minimal polynomial.
Argument. Since $\omega^3 = 1$ and $\omega \neq 1$, the roots are $a, a\omega, a\omega^2$. Thus $K = \mathbb{Q}(a, \omega)$ contains all roots. Conversely a subfield containing all roots contains a and $\frac{a\omega}{a} = \omega$, so must contain K. The polynomial $x^3 - 2$ is Eisenstein at two, giving $[\mathbb{Q}(a) : \mathbb{Q}] = 3$. Every element of $\mathbb{Q}(a)$ is real. Neither ω nor its conjugate is real, so $x^2 + x + 1$ has no root there and is irreducible over that field. Hence $[K : \mathbb{Q}(a)] = 2$, and the tower gives degree six.
A tempting dead end. Three roots do not imply degree three for their joint field. The real root alone does not supply the nonreal roots.
Hypotheses. The argument uses a specified embedding into the complex numbers and the reality of the first field. Nonzero a makes the root ratio legitimate.
Check your argument. Prove both containments for the smallest field and justify irreducibility at both stages.

8 Unit 8 · Finite fields and applications

MM5020-U8.01 · E · 15 min
Learning objective: U8.1 Field orders

Which of the orders $8, 9, 10, 16, 18, 25$ can occur for a finite field?

The possible orders are $8 = 2^3$, $9 = 3^2$, $16 = 2^4$, and $25 = 5^2$. Fields of each exist. Ten and eighteen involve distinct primes and are not prime powers, so neither occurs.

Check your argument. State both necessity and existence.

MM5020-U8.02 · E · 15 min
Learning objective: U8.1 Arithmetic in four elements

In $\mathbb{F}_4 = \mathbb{F}_2[\alpha]$ with $\alpha^2 + \alpha + 1 = 0$, compute $\alpha(\alpha + 1)$ and the inverse of $\alpha + 1$.

$\alpha(\alpha + 1) = \alpha^2 + \alpha = (\alpha + 1) + \alpha = 1$. Thus the inverse of $\alpha + 1$ is α.

Check your argument. Use characteristic two correctly.

MM5020-U8.03 · E · 15 min
Learning objective: U8.2 Generators

Find the order of two in $\mathbb{F}_7^\times$.

The powers are $2, 4, 1$, so the order is three. Two is therefore not a generator of the six-element multiplicative group.

Check your argument. Check that no earlier positive power is one.

MM5020-U8.04 · E · 15 min
Learning objective: U8.1 Characteristic and dimension

What are the characteristic and dimension over the prime field of a field of order eighty-one?

The characteristic is three and the dimension over $\mathbb{F}_3$ is four, because a d-dimensional vector space over $\mathbb{F}_3$ has 3^d elements.

Check your argument. Distinguish characteristic from total cardinality.

MM5020-U8.05 · C · 30 min
Learning objective: U8.2 Frobenius

Prove the map $a \mapsto a^p$ is an automorphism of a finite field of characteristic p.

In characteristic p, binomial coefficients make $(a + b)^p = a^p + b^p$; multiplication and one are also preserved. If $a^p = b^p$, then $(a - b)^p = 0$, so $a = b$. The field is finite, so injectivity implies surjectivity. Thus the map is an automorphism.

Check your argument. State where finiteness is used; Frobenius need not be surjective in an infinite field.

MM5020-U8.06 · C · 30 min
Learning objective: U8.2 Interpolation

Find the degree-less-than-three polynomial over $\mathbb{F}_5$ with $f(0) = 1$, $f(1) = 0$, $f(2) = 2$.

Write $f(x) = ax^2 + bx + 1$ using the value at zero. The other two conditions give $a + b = 4$ and $4a + 2b = 1$ modulo five. Subtract twice the first equation to get $2a = 3$, hence $a = 4$ and $b = 0$. Thus $f(x) = 4x^2 + 1$. Substitution checks all three values; uniqueness follows from the root bound.

Check your argument. Use modular inverses and explain uniqueness.

MM5020-U8.07 · C · 30 min
Learning objective: U8.1 Polynomial roots

Show every element of $\mathbb{F}_q$ is a root of $x^q - x$, and this polynomial has no repeated root.

Zero is a root. Every nonzero a belongs to a group of order $q - 1$, so $a^{q-1} = 1$ and $a^q = a$. Since q is zero in the field, the derivative is $qx^{q-1} - 1 = -1$, which shares no root with the polynomial, so no root repeats.

Check your argument. Handle zero separately and use characteristic correctly in the derivative.

MM5020-U8.08 · C · 30 min
Learning objective: U8.2 Evaluation codes

Evaluate all degree-less-than-two polynomials at the five elements of $\mathbb{F}_5$. How many codewords result, and what is their minimum distance?

There are $5^2 = 25$ polynomials, and evaluation is injective because a nonzero degree-at-most-one polynomial cannot vanish at five points. Two distinct codewords differ in at least four coordinates. This bound is attained by the polynomials zero and x, whose evaluations agree only at zero. Thus the minimum distance is four.

Check your argument. Show the bound is attained, not only that it is a lower bound.

MM5020-U8.09 · C · 30 min
Learning objective: U8.2 Primitive elements

Find all generators of $\mathbb{F}_7^\times$, given that three is one generator.

The generators are $3^1 = 3$ and $3^5 = 5$ modulo seven. The order formula shows no other powers generate because their exponents share a nontrivial gcd with six.

Check your argument. Explain exhaustiveness using the cyclic-group order formula.

MM5020-U8.10 · A · 50 min
Learning objective: U8.1 Subfields

Show a field of sixteen elements has a unique subfield of four elements.

The cyclic multiplicative group of order fifteen has exactly three solutions to $x^3 = 1$: for a generator g, the exponents satisfy $15|3k$, giving $k = 0, 5, 10$. Including zero gives four roots of $x^4 - x$. The fourth-power Frobenius identity shows these roots are closed under addition and multiplication, and nonzero inverses also remain roots, so they form a subfield. Any four-element subfield consists entirely of such roots and hence equals this one.

Check your argument. Prove existence by closure and uniqueness by the root characterization.

MM5020-U8.11 · A · 50 min
Learning objective: U8.2 Polynomial functions

Prove every function $\mathbb{F}_q \to \mathbb{F}_q$ is represented by a unique polynomial of degree less than q. Is the representing polynomial unique without the degree bound?

Interpolation gives a polynomial of degree less than q with any prescribed function values, and the root bound gives uniqueness in that degree range. Without the bound, adding any multiple of $x^q - x$ changes the formal polynomial without changing any value on the field. Thus uniqueness fails among all polynomials.

Check your argument. Distinguish formal polynomials from the functions they induce.

MM5020-U8.12 · A · 50 min
Learning objective: U8.1 Irreducible quadratics

Count the monic irreducible quadratics over $\mathbb{F}_q$.

Decisive idea. Use unique factorization to replace reducible monic quadratics by unordered root multisets.

Finding the approach. Count the complement, where reducibility has a concrete description by roots. Separate repeated roots from distinct roots to avoid ordering errors.

Argument. There are q^2 monic quadratics. A reducible one has the form $(x-a)(x-b)$ with $a, b \in \mathbb{F}_q$, and unique factorization identifies it with an unordered pair with repetition. There are $\frac{q(q+1)}{2}$ such pairs. Thus the number of monic irreducible quadratics is $q^2 - \frac{q(q+1)}{2} = \frac{q(q-1)}{2}$.

There are q repeated-root polynomials $(x-a)^2$. There are $\frac{q(q-1)}{2}$ pairs of distinct roots. Unique factorization ensures that no two different unordered multisets produce the same monic polynomial. Thus the reducible count is $q + \frac{q(q-1)}{2}$, giving the claimed remainder. For $q = 2$, the one irreducible example is $x^2 + x + 1$, which checks the formula.

A tempting dead end. Counting ordered pairs gives each distinct-root polynomial twice but each repeated-root polynomial once. Dividing all q^2 ordered pairs by two therefore gives the wrong answer.

Hypotheses. Over a field, a quadratic is reducible exactly when it is a product of two linear factors. The count remains valid in characteristic two; no division by two inside the field is being performed.

Check your argument. Include repeated roots and justify why no reducible quadratic is missed or counted twice.

MM5020-U8.13 · A · 60 min · Optional challenge

Learning objective: U8 Transfer: An equation over a larger finite field

For each $b \in \mathbb{F}_q$, determine the number of $x \in \mathbb{F}_{q^2}$ satisfying $x^q + x = b$. Include the case of characteristic two.

Decisive idea. Identify the equation as a nonzero linear functional, then count all fibers from its kernel.

Finding the approach. The equation is a fiber of a map between finite vector spaces. This avoids trying to solve a degree-q polynomial separately for each value of b.

Argument. Define $T(x) = x^q + x$. Since $x^{q^2} = x$, we have $T(x)^q = x^{q^2} + x^q = T(x)$, so $T(x)$ lies in $\mathbb{F}_q$. Frobenius additivity and $c^q = c$ for $c \in \mathbb{F}_q$ show that T is $\mathbb{F}_q$-linear. It is not the zero map: the nonzero polynomial $X^q + X$ of degree q cannot vanish at all q^2 elements. A nonzero linear map to a one-dimensional space is surjective. Its domain has dimension two, so its kernel has dimension one and cardinality q. Every fiber is a translate of the kernel; consequently there are exactly q solutions for every b.

A tempting dead end. In odd characteristic $T(1) = 2$ proves nonzeroness, but that argument fails in characteristic two. A failed test vector does not mean the linear map is zero.

Hypotheses. Use the field $\mathbb{F}_{q^2}$ with its subfield of q elements and the polynomial root bound over a field. The argument avoids division by two.

Check your argument. Prove the codomain, linearity, nonzeroness in every characteristic, and the fiber count.

9 Mock assessment 1

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM5020-M1.01 · E · 15 min
Learning objective: U1–U2 Cyclic groups
List the orders of every element of $\mathbb{Z}/8\mathbb{Z}$.
For $k = 0, 1, 2, 3, 4, 5, 6, 7$, the orders are respectively $1, 8, 4, 8, 2, 8, 4, 8$. The formula follows by asking for the least positive t with $8 | kt$.
Check your argument. Explain the formula and treat the zero residue.

MM5020-M1.02 · E · 15 min
Learning objective: U5 Ring quotients
Determine whether $\mathbb{Z}/15\mathbb{Z}$ is a field and list its units.
It is not a field, since nonzero $[3]$, $[5]$ multiply to zero. Its units are $[1]$, $[2]$, $[4]$, $[7]$, $[8]$, $[11]$, $[13]$, $[14]$, exactly the residues coprime to fifteen by Bézout.
Check your argument. Justify the full unit list.

MM5020-M1.03 · C · 30 min
Learning objective: U3–U4 Group structure
Let $|G| = 49$. Prove G is abelian, and show that if some element has order forty-nine then G is cyclic.
The center is nontrivial. If it had order seven, the quotient by the center would be cyclic of order seven, forcing G abelian and contradicting that center size. Thus the center is all of G. An element of order forty-nine has a generated subgroup of that size, equal to G.
Check your argument. The cyclicity conclusion is conditional; abelian does not automatically mean cyclic.

MM5020-M1.04 · C · 30 min
Learning objective: U6–U7 Algebraic extensions
Let $\alpha^3 = 3$ over $\mathbb{Q}$. Prove the extension degree is three and find $(\alpha - 1)^{-1}$.
$x^3 - 3$ is Eisenstein at three, so it is the minimal polynomial and the degree is three. Since $(\alpha - 1)(\alpha^2 + \alpha + 1) = \alpha^3 - 1 = 2$, the inverse is $\frac{\alpha^2+\alpha+1}{2}$.
Check your argument. Prove irreducibility and verify the inverse.

MM5020-M1.05 · A · 40 min
Learning objective: U3 Universal constructions
For normal subgroups N, M of G, show $g \mapsto (gN, gM)$ induces an injective homomorphism $G/(N \cap M) \to G/N \times G/M$. When is it surjective?
The paired map is a homomorphism with kernel $N \cap M$, so its induced map is injective by the first isomorphism theorem. If it is surjective, any pair (aN, M) has a lift $g \in M$ with $aN = gN$, so $a \in MN = NM$. Conversely if $G = NM$, for prescribed aN, bM write $b^{-1}a = mn$ with $m \in M, n \in N$. Then $g = bm = an^{-1}$ has both prescribed cosets.
Check your argument. Give the kernel and prove both directions of the surjectivity criterion.

MM5020-M1.06 · A · 40 min
Learning objective: U8 Field structure
Prove that the fixed field of Frobenius $x \mapsto x^p$ in $\mathbb{F}_{p^n}$ is exactly $\mathbb{F}_p$.
Frobenius fixes the prime field because it fixes one and preserves addition. Thus its fixed set contains p elements. Any fixed element is a root of the degree-p polynomial $x^p - x$, which has at most p roots in a field. Hence there are no others and the fixed field is $\mathbb{F}_p$.
Check your argument. Show both containment and the cardinality bound.

10 Mock assessment 2

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM5020-M2.01 · E · 15 min
Learning objective: U2 Permutations
Find the order and sign of $(123)(45)(67)$.
The order is six. Its sign is $(-1)^2(-1)(-1) = 1$, so it is even.
Check your argument. Distinguish order from parity.

MM5020-M2.02 · E · 15 min
Learning objective: U7 Minimal polynomials
Find the minimal polynomial of $2 + \sqrt{5}$ over $\mathbb{Q}$.
The polynomial is $x^2 - 4x - 1$. It vanishes at the given element and has roots $2 \pm \sqrt{5}$, neither rational, so it is irreducible and minimal.
Check your argument. Verify irreducibility.

MM5020-M2.03 · C · 30 min
Learning objective: U3 Quotient isomorphisms
Prove $S_4/A_4 \cong C_2$.
Sign is a homomorphism onto $\{1, -1\}$: the identity maps to one and any transposition to negative one. Its kernel is A_4. The first isomorphism theorem gives the quotient isomorphism, and the target is cyclic of order two.
Check your argument. Establish surjectivity and the kernel.

MM5020-M2.04 · C · 30 min
Learning objective: U5–U6 Field construction
Prove $\mathbb{F}_3[x]/(x^2 + 1)$ is a field and find the inverse of $[x + 1]$.
The values of $x^2 + 1$ at zero, one, and two are one, two, and two, so the quadratic is irreducible. The quotient is a field. Since $(x + 1)(x - 1) = x^2 - 1 = -2 = 1$ in the quotient, the inverse is $[x - 1]$.
Check your argument. Use arithmetic in the specified characteristic.

MM5020-M2.05 · A · 40 min
Learning objective: U4 Actions
A group G of order fifteen acts on a set of seven elements. Prove there is a fixed point.
Every orbit size divides fifteen and is at most seven, so it is one, three, or five. If no fixed point existed, seven would be a sum of threes and fives. With at most one five, the remaining two is impossible; with no five, seven is not divisible by three. Hence an orbit of size one exists, which is a point fixed by all of G.
Check your argument. A fixed point means fixed by the entire group, not only one element.

MM5020-M2.06 · A · 40 min
Learning objective: U7–U8 Degrees and subfields
Prove that $\mathbb{F}_{p^3}$ has no subfield strictly between $\mathbb{F}_p$ and itself.
For an intermediate field F, the tower law gives $3 = \left[\mathbb{F}_{p^3} : F\right]\left[F : \mathbb{F}_p\right]$. Both factors are positive integers, so one is one. If $\left[F : \mathbb{F}_p\right] = 1$, then $F = \mathbb{F}_p$; otherwise $\left[\mathbb{F}_{p^3} : F\right] = 1$ and F is the whole field.
Check your argument. Explain why degree one forces equality of fields.